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SOVEREIGN INSIGHT ANALYSIS

Project Documentation

Currency Exchange Rate Analytics System

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1. General Information

Primary Topic: Statistical evaluation of foreign exchange rate fluctuations using the official NBP API endpoints.

Core Objective: Design and engineer a web application that empowers users to conduct statistical assessments and cross-reference various currency exchange rates, relying exclusively on data provided by the National Bank of Poland (NBP).

Development Methodology: The project lifecycle is governed by Agile/SCRUM principles, incorporates Continuous Integration/Continuous Deployment (CI/CD) pipelines, and utilises Git for collaborative version control.

Application Name: Sovereign Insight Analysis
Backlog / Task Tracking: GitHub Projects
Frontend: Framework React 19 (Vite, JSX)
Styling: TailwindCSS
CI/CD: GitHub Actions
Data Source: NBP Web API — http://api.nbp.pl/

2. System Architecture

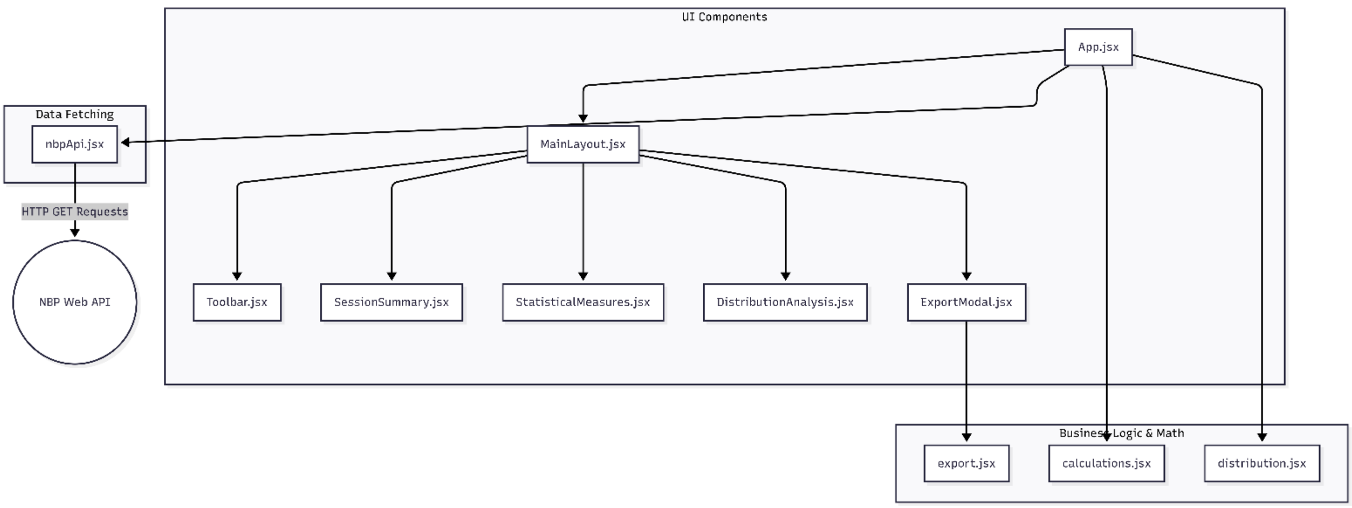
2.1 Architecture Description

Sovereign Insight Analysis is designed as a modern Single Page Application (SPA) built with the React.js framework (v19) and bundled with Vite. The architecture follows a client-side processing model: the frontend communicates directly with the NBP Web API, with all business logic — statistical calculations, cross-rate evaluation, request chunking, and data aggregation — executed entirely within the user's browser. This approach eliminates backend infrastructure dependencies, ensures real-time processing, and maximises performance.

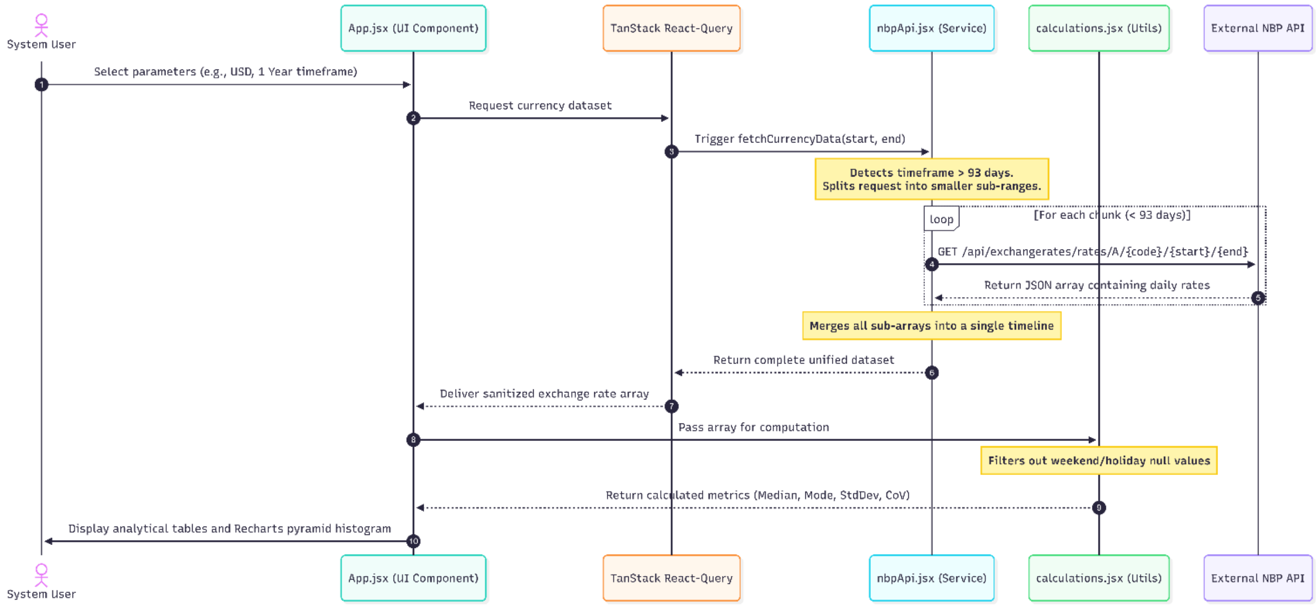
The codebase is organised into three main layers:

- UI Layer (React Components): App.jsx, layout components (MainLayout, Toolbar, SessionSummary, StatisticalMeasures, DistributionAnalysis, ExportModal).
- Service Layer: nbpApi.jsx — handles all communication with the NBP API, including 93-day chunking and result merging.
- Utility Layer: calculations.jsx (session logic, cross-rates, statistics), distribution.jsx (14-bin histogram logic), export.jsx (CSV/JSON/image export).

2.2 Component Diagram



2.3 Sequence Diagram — Data Fetching & Statistics



2.4 Activity Diagram — User Analysis Flow